CLASS 9 — MATHS · FORMULA SHEET

Chapter 8: **Sequences and Progressions** | Quick revision, one page

Sequences

- A sequence is an ordered list of numbers $t_1, t_2, t_3, \ldots$; t_n is the nth term.
- A rule such as $t_n = 3n + 2$ gives every term by putting $n = 1, 2, 3, \ldots$

Arithmetic progression (AP)

- Common difference $d = t_{n+1} - t_n$ is constant.
- nth term: $t_n = a + (n-1)d$.
- Sum of first n terms: $S_n = \frac{n}{2}\,[2a + (n-1)d] = \frac{n}{2}(a + l)$, where l is the last term.
- $t_n = S_n - S_{n-1}$ for $n \geq 2$.
- Sum of first n natural numbers: $\frac{n(n+1)}{2}$.
- Three numbers $a - d,\ a,\ a + d$ are in AP; the middle term is the average of its neighbours.

Geometric progression (GP)

- Common ratio $r = \frac{t_{n+1}}{t_n}$ is constant (and non-zero).
- nth term: $t_n = ar^{n-1}$.
- Terms are multiplied by r each step; $|r| < 1$ makes the terms shrink, $r < 0$ makes the signs alternate.

Telling them apart

- AP: add the same number each time (constant difference). GP: multiply by the same number each time (constant ratio).
- Example: $2, 5, 8, \ldots$ is an AP; $2, 6, 18, \ldots$ is a GP; $1, 4, 9, 16, \ldots$ is neither.

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